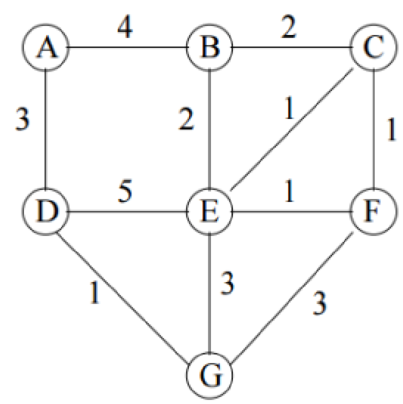
INT202 Complexity of Algorithms
Tutorial 5

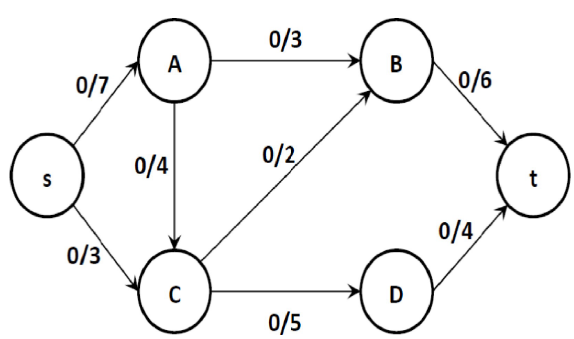
1. Prim and Kruskal Executions

1) Execute Prim's algorithm on the following graph starting at vertex A. If there are any ties, the vertex with the lower letter comes first. List the edges in the order in which they are added to the tree.

2) Execute Kruskal's algorithm on the following graph starting at vertex A. Assume that equal weight edges are ordered lexicographically by the labels of their vertices assuming that the lower labeled vertex always comes first when specifying an edge, e.g. (C, E) is before (C, F) which in turn is before (D, G). List the edges in the order in which they are added to the developing forest.



2. Execute the Ford-Fulkerson maximum flow algorithm on the following network:



(1) For each step of the algorithm, write in a table the augmenting path by listing its vertices, its residual capacity, and the resulting flow on the network.

(2) What is the value of the maximum flow?

(3) List the vertices on the s side of the minimum cut.

(4) What is the capacity of the minimum cut?